

Geography Ch-12 (Advanced) — The Indian Ocean: Monsoon Currents & the IOD

Advanced / India-applied | GS-I & GS-III | Source: GC Leong + Majid Husain (Indian Ocean circulation)
North Indian Ocean seasonal current reversal; South Indian Ocean gyre; salinity contrasts
Extra CA: Indian Ocean Dipole (distinct from Must-Do's AMOC CA)

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

ADVANCED — Indian Ocean Circulation & Monsoon Reversal (GC Leong / Majid)

■ NEWS TRIGGER

The North Indian Ocean is unique — its surface currents COMPLETELY REVERSE with the monsoon, unlike the steady gyres of other oceans, tightly coupling the sea to India's rainfall.

■ INTRO & ORIGIN

GC Leong: in the North Indian Ocean there is a **complete reversal** of current direction between summer and winter, driven by the changing monsoon winds; the South Indian Ocean has a normal gyre.

*Because the monsoon wind flips direction seasonally, the wind-driven surface current flips with it — a textbook demonstration that **wind drives surface currents**.*

■ KEY DATA

Region / season	Current	Direction / note
N Indian Ocean, summer (Jun-Oct)	SW Monsoon Drift	Blown from the south-west (clockwise)
N Indian Ocean, winter (from Dec)	NE Monsoon Drift	Blown from the north-east (reversed)
S Indian Ocean	South Equatorial Current	Feeds the gyre
S Indian Ocean (W limb)	Agulhas / Mozambique	Warm current past Madagascar
S Indian Ocean (S/E limb)	West Wind Drift -> W. Australian	Completes the circuit

■ STATIC THEORY

- **North Indian Ocean:** a complete seasonal reversal — the **South-West Monsoon Drift** (summer, Jun-Oct) reverses to the **North-East Monsoon Drift** (winter, from Dec) as the monsoon wind flips.
- This is the clearest natural proof that **wind is the dominant driver** of surface currents (GC Leong).
- **South Indian Ocean:** a normal subtropical gyre — **South Equatorial Current -> Agulhas/Mozambique (warm, past Madagascar) -> West Wind Drift -> West Australian Current**.
- **Salinity contrast:** the **Arabian Sea** is more saline (high evaporation, few rivers) than the **Bay of Bengal** (diluted by the Ganga-Brahmaputra and heavy rain).
- Seasonal upwelling off Somalia/west India (SW monsoon) boosts nutrients and fisheries.

■ MUST-KNOW FACTS

1. N Indian Ocean currents REVERSE seasonally with the monsoon.
2. Summer = SW Monsoon Drift; winter = NE Monsoon Drift.
3. S Indian Ocean gyre: S. Equatorial -> Agulhas/Mozambique -> West Wind Drift -> W. Australian.
4. Arabian Sea saltier than Bay of Bengal (evaporation vs river dilution).

■ UPSC TRAPS

- ✗ North Indian Ocean currents flow in a fixed gyre all year.
- ✓ They COMPLETELY REVERSE between summer (SW) and winter (NE) monsoons.
- ✗ The Bay of Bengal is more saline than the Arabian Sea.
- ✓ The ARABIAN SEA is more saline (high evaporation, few rivers); the Bay is diluted by big rivers.

5. Proves wind is the dominant driver of surface currents.

■ MAINS ANGLE

Explain why the North Indian Ocean's currents reverse seasonally and how Arabian Sea vs Bay of Bengal salinity and upwelling shape India's fisheries and climate.

■ STUDY LINK

Geography → India → Indian Ocean Circulation (GC Leong: Indian Ocean; Majid: Currents)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA ANCHOR — The Indian Ocean Dipole (IOD)

GS-I / GS-III ·
Indian Ocean
applied →
Monsoon
variability

■ NEWS TRIGGER

The Indian Ocean Dipole (IOD) — the see-saw of sea-surface temperature between the western and eastern Indian Ocean — is a key modulator of the Indian monsoon, watched every year alongside ENSO.

■ INTRO & ORIGIN

IOD is an ocean-atmosphere coupling built on the same SST/salinity physics: a warm-west/cool-east pattern (positive IOD) tends to strengthen India's monsoon.

■ STATIC THEORY

- **Positive IOD:** warmer western Indian Ocean, cooler east -> tends to **strengthen the Indian monsoon** (dry Indonesia/Australia).
- **Negative IOD:** cooler west, warmer east -> tends to **weaken the monsoon** (wet Indonesia/Australia).
- IOD often reinforces or offsets **ENSO**; climate change is altering its behaviour and predictability.

■ MUST-KNOW FACTS

1. IOD = west-vs-east Indian Ocean SST difference.
2. Positive IOD -> stronger Indian monsoon; Negative -> weaker.
3. Works alongside ENSO to modulate monsoon rainfall.

■ UPSC TRAPS

- ✗ A positive IOD weakens the Indian monsoon.
- ✓ A POSITIVE IOD (warm west) tends to STRENGTHEN the Indian monsoon.

■ MAINS ANGLE

Link Indian Ocean SST physics (IOD) to monsoon forecasting and agrarian risk.

■ STUDY LINK

Geography → Indian Ocean -> IOD & Monsoon (applied CA)

End of Register Notes — UPSC Agent / Copilot CLI